

SELEHADIN NASIR

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Education

Addis Ababa University (AAU)

Bachelor of Science in Electrical and Computer Engineering

Addis Ababa, Ethiopia

Expected Graduation: May 2028

Projects

BOOKR | *React, Node.js, Express, PostgreSQL, Telegram Bot API (grammY)*

- Built and deployed a bilingual (Amharic/English) Telegram mini-app for service booking, live in production on Render/Supabase, currently onboarding local businesses in Addis Ababa.
- Designed a backtracking slot-search algorithm with radius-bounded expansion and gap-minimization scoring, plus a shared-resource capacity model and sequential dependent-service scheduling across different staff members.
- Implemented transactional booking and cancellation flows with row-level locking (PostgreSQL `FOR UPDATE`) to prevent race conditions under concurrent requests.
- Built a full-featured Telegram bot (grammY) handling multilingual notifications, staff onboarding, and automated booking-resolution reminders.

CodeDash | *React, Node.js, Express, MongoDB, Socket.io, Redis, Bull, JWT*

- Engineered a custom VS Code extension (VS Code API) that monitors local active files, keystrokes, and session times, utilizing a local offline queue to buffer data during network drops.
- Developed a scalable Node.js/Express backend that receives buffered session payloads and syncs active workspace data over WebSockets (Socket.io) to minimize API overhead.
- Integrated with the GitHub API to fetch and aggregate historical repository data, including commits, programming languages, and user activity metrics.
- Implemented background task queues using Bull and Redis to handle asynchronous data synchronization without blocking the main event loop.

Smart Logic Simulator | *Java, JavaFX, CSS, Object I/O Streams, SVGPath*

- Developed an interactive digital logic simulator in Java and JavaFX, separating UI rendering from underlying logic models, with a custom real-time simulation engine driven by a high-frequency `AnimationTimer` clock.
- Designed custom binary serialization (*.ckt) using Object I/O streams and dynamic, vector-rendered connection wires via `CubicCurve` mathematical modeling.

Induction Motor Fault Detector | *Python, TensorFlow, scikit-learn, NumPy, Tkinter*

- Applied FFT to decompose motor current signals into the frequency domain, extracting harmonic amplitudes as diagnostic features for fault classification.
- Trained a Random Forest classifier and a parallel 1D CNN on synthetic motor-signal data, achieving 100% accuracy on held-out test data, with a Tkinter GUI for non-technical users to visualize diagnoses.

Certifications

Introduction to AI — University of Helsinki & MinnaLearn (Elements of AI)

Technical Skills

Languages: JavaScript, Java, Python

Web Technologies: React, Node.js, Express, MongoDB, PostgreSQL, Socket.io, JWT, HTML5/CSS3, Tailwind CSS

Hardware & Tools: VS Code Extension API, Git, GitHub, Vercel, Netlify, Render, Supabase, Bull, Redis, Logisim-Evolution

Familiar with: TensorFlow, scikit-learn, NumPy, Verilog, C, MATLAB